Subject:	Database Development 171/181
Assessment:	Project
Total:	100 Marks

Introduction

A database that meets all user requirements requires careful planning, design and implementation. In this assignment, you are expected to demonstrate the knowledge and skills you have learned throughout this course by developing a **Microsoft Access database** that meets specific user requirements as outlined in this document.

The assignment is divided into two parts. The objective of Section 1 is to create complete, normalized, non-redundant, and fully integrated database design that meets the user requirements. In Section 2, you create the database in Microsoft Access™ database management software, load data into the database, and create all necessary objects that enable end-users to work with the databases without any assistance.

Section 1: Database Design

A small community fitness centre called *ActiveLife Fitness* currently manages its operations using paper forms and Excel spreadsheets. The centre currently has:

- Around 150 active members and still growing
- 5 fitness trainers and other staff including receptionists and administrative employees
- Group classes (e.g., Yoga, Spinning, HIIT, Aerobics, Zumba Dance, Pilates, etc)
- Monthly membership payments with an option for walk-in clients

The business owner wants a database to manage memberships, trainers, classes, and payments more efficiently. The owner developed a list of business rules and specifications to begin the design of an Entity Relationship (ER) model:

1. **Memberships:**
   - A member may have an account but there is an option for cash payments.
   - Each account is linked to one member only (no joint accounts).
   - Members may sign up for multiple classes.
   - Each member has: Membership ID, Name, Address, City, Province, Telephone, Date of Birth, and Membership Type (e.g., Cash, Monthly, Annual).
   - For members with personal training accounts, store the account number, type of account and date opened. Also track the track balance, last payment date, last payment amount, and whether it's closed or not.
2. **Employees:**
   - Staff include trainers, receptionists, cashiers and other administrative employees.
   - Each employee works in one role/position and may report to another employee (e.g. supervisor, senior trainer)
   - Track the Employee ID, Names, Address, Telephone, Date of Hire, Position, Salary, Date of Birth, and Specialization/Skill.

- Trainers may work on part-time or full-time contract.
- Trainers may teach one or more classes.
- Non-trainer employees do not have assigned classes.

3. **Classes:**
   - Classes have: Class ID, Description, Class Type (e.g., Yoga, HIIT, Pilates), Schedule, Duration, and Maximum Capacity.
   - Classes can have multiple members enrolled.
   - Classes are taught by one trainer (one-to-many from trainer to class).
   - Classes are conducted in various rooms.
   - Rooms have a RoomID, Name, Location, Capacity
   - Each room can host multiple classes, but each class is linked to only one room.
4. **Bookings:**
   - Members book classes through a reception desk (or online).
   - Track: Booking ID, Member ID, Class ID, Booking Date, and Payment Status.
5. **Payments:**
   - Track: Payment ID, Member ID, Cashier, Amount, Date, and Payment Method.
   - Payments may be linked to membership fees or personal training sessions.
6. **Equipment:**
   - Each class may require equipment.
   - Equipment has: Equipment ID, Description, Brand name, Purchase Date, Cost, and Quantity Available.

Based upon this information, draw a detailed Entity Relationship Diagram (ERD) that includes the identified entities, attributes and clearly labelled relationships. Your diagram should have a way of identifying all primary keys and foreign keys clearly. You may discover more entities and decomposable attributes from the above rules. Please indicate any additional assumptions that you have made.

NB: Marks for documentation or presentation will be awarded where the appropriate software tools were used to design the ERD. Using pen and paper is not allowed, neither is the use of Microsoft Access as a design tool. All tables should be normalized to at least 3NF.

Section 2: Microsoft Access

You have planned your system in Section 1, now it is time to implement the database in Microsoft Access database management software. When finished you will present the final database application.

The system requirements are as follows:

1. Follow your database design to create the tables in the database.
2. Be certain to specify the field names, data types, field descriptions, and field sizes as you create each table.

3. Create validation rules and validation text, input masks, fields captions, default values, and lookup fields as necessary to ensure the integrity of data stored in the tables.
4. Create relationships between all the tables. Enforce referential integrity in each relationship.
5. After you have created all your tables, populate them with some records for testing.
6. Develop at least five (5) queries for use by the business. The queries should answer questions such as:
    - Which members are enrolled in the "Yoga" class this week?
    - What is the total income received for each month per class type (e.g., Yoga)?
7. Create user-friendly forms for data entry, navigation, and display purposes.
8. Create reports which the staff can print and distribute to people who need them.
9. One of the forms should serve as the main form to navigate through the database application. Consider using Switchboards, Navigation Forms, or other custom menus.
10. Demonstrate your knowledge of macros by implementing some basic user interactions.
11. Add a splash screen that runs for three seconds, after which it continues to the database.

NB: The queries, forms and reports are any of your own choosing; however, the logical reasoning behind your choice must be sound. An advanced query is when it has additional criteria needed to execute or the data spans over more than one table.

Mark Allocation

Content	
SECTION 1:	**Subtotal: 40**
Project Outline	**05**
ER Diagram	**30**
- Entities	10
- Relationships, PKs and FKs	10
- Entity Attributes	10
Documentation (Layout, Presentation & Justifications)	**5**
SECTION 2:	**Subtotal: 60**
Tables and Relationships (created as per ERD with sample data)	**10**
Forms and Reports	**25**
- Forms *(x3 minimum + 4 marks for Layout and Design)*	10
- Reports *(x3 minimum + 4 marks for Layout and Design)*	10
- Navigation (Main Menu, Switchboard, Navigation Forms, etc.)	5
Queries	**15**
- Basic Queries (Select, Append, Update, Delete, Simple Joins)	5
- Advanced Queries (MakeTable, Totals, Crosstab, Expressions, Multi-table joins, etc.)	10
Splash Screen, macros and other extra effort	**5**
Overall Presentation of Database Application	**5**
Total:	**100**

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Additional Information

- This is a group assignment. Groups may not exceed 4 people.
- All work must be original. Copying or reproducing someone's work will not be tolerated.
- Submit both the documentation and the working Access database application Moodle (BC Connect) before the due date.
- Marks will be awarded for both content and presentation skills.
- A cover page, table of contents and bibliography (or reference list) must be included.
- The cover page should include names of all group members who participated in the project.
- List all your sources of information (book titles, website addresses, journals, etc.) on the last page of your presentation under the headline "References".
- Belgium Campus makes use of software that can **scan for plagiarism** and penalties apply for any plagiarised work including getting a 0 for this assignment
- No mark will be awarded if the project is not uploaded via BC Connect.
- Late submissions will not be accepted; missing the deadline is an automatic 0.